gpeyt1-43 · Eccentric footing (Meyerhof)

GEO PE · Long Calculation Problems I — Topic 9: Shallow Foundations

Problem

An 8 ft × 8 ft footing supports a vertical load P = 180 kips (including footing weight) and an overturning moment M = 120 kip-ft about one axis from wind on a sign structure. The footing bears on dense sand where settlement analysis uses the Meyerhof equivalent uniform pressure. Allowable bearing pressure is 4.0 ksf.

Most nearly, the Meyerhof equivalent uniform bearing pressure is:

A. 4,219 psf
B. 2,812 psf
C. 3,375 psf
D. 3,068 psf

Given equations / data

> $e = M/P \ (\le B/6)$; $B' = B - 2e$; $q_{eq} = P/(B'L)$; $q_{max} = (P/BL)(1 + 6e/B)$

Worked solution

1. $e = 120/180 = 0.667$ ft $\le B/6 = 1.33$ ✓
2. $B' = 8 - 2(0.667) = 6.667$ ft
3. $q_{eq} = 180/(6.667 \times 8) = 3.375$ ksf $= 3{,}375$ psf
4. ($q_{max} = 4{,}219$ psf)

Final answer

3,375 psf

Why the other values are wrong

- **4,219 psf** — reported qmax (trapezoidal toe pressure)
- **2,812 psf** — ignored eccentricity
- **3,068 psf** — used $B' = B - e$

References (Jayce's library)

- NCEES PE Civil Reference Handbook v2.1 Applied stress at base — $q_{eq} = \Sigma V/B'$, $B' = B - 2e$ (cantilever wall figure, p. 209); NHI-06-089 §8 effective width

Original problem inspired by the problem type in: https://www.youtube.com/watch?v=du7B1MO-GCg (no creator values copied).

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